`stats_v_process.ipynb` — Notebook Documentation

SLR Forecasting Project

March 2026

1 Purpose

This notebook compares the forecasting skill of three approaches to projecting global mean sea level rise (GMSLR):

1. A **Bayesian rate-and-state semi-empirical model** ("DOLS") that relates GMSLR rate to global mean surface temperature (GMST) via a quadratic polynomial with a thermal disequilibrium state variable;
2. **IPCC AR6 process-model projections**, comprising seven workflows built on ISMIP6 ice-sheet models, glacier models, and thermosteric components;
3. The **Vermeer & Rahmstorf (2009)** dual semi-empirical model ("VR09"), $dH/dt = a(T - T_0) + b\,dT/dt$, included as a second semi-empirical benchmark.

The comparison is carried out using out-of-sample verification against NASA satellite altimetry (2006–2024), hindcast cross-validation with truncated calibration records, and several forecast-horizon diagnostics.

2 Notebook Structure

The notebook contains 30 cells organized into 8 sections.

2.1 Section 1: Data Loading and Baseline Alignment (Cells 1–3)

All datasets are loaded from the project HDF5 store and rebased so that the year 2000 corresponds to zero sea level and zero temperature anomaly.

Datasets loaded:

- Frederikse et al. tide-gauge GMSL reconstruction (1900–2018, annual, meters)
- Berkeley Earth GMST (1850–2024, annual, with 10-year and 20-year Lowess smoothing)
- NASA satellite altimetry GMSL (1993–2024, monthly)
- IPCC AR6 GMSL projections for SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5 (medium confidence, quantile summaries)
- IPCC AR6 GMST projections for matching SSPs

Baseline alignment: IPCC temperatures are shifted from their pre-industrial baseline to the Berkeley Earth 1995–2005 baseline by matching the overlap at 2015. GMSL is rebased to year 2000 = 0 for all sources.

2.2 Section 2: Bayesian Rate-and-State Model (Cells 4–6)

The calibrated Bayesian rate-and-state model is loaded from `bayesian_ratestate_results.json`. The model expresses the GMSLR rate as

$$\frac{dH}{dt} = \frac{d\alpha}{dT}\,T^2 + \alpha_0\,T + c + d\,(S - T), \tag{1}$$

where T is GMST (Berkeley Earth 1995–2005 baseline), c is a background trend, and $S(t)$ is a slow thermal state variable governed by

$$\tau\,\frac{dS}{dt} = T - S, \tag{2}$$

with relaxation timescale τ. The term $d\,(S - T)$ represents thermal disequilibrium between fast surface warming and slow deep-ocean/ice-sheet adjustment.

Calibrated coefficients:

Parameter	Value	Unit
$d\alpha/dT$	4.625	$\mathrm{mm\,yr^{-1}\,{}^\circ C^{-2}}$
α_0	5.171	$\mathrm{mm\,yr^{-1}\,{}^\circ C^{-1}}$
c (trend)	2.632	$\mathrm{mm\,yr^{-1}}$
d (diseq.)	0.532	$\mathrm{mm\,yr^{-1}\,{}^\circ C^{-1}}$
τ	18.2	yr
R^2	0.635	—

Ensemble projections are generated via Monte Carlo sampling ($N = 2000$) from the joint posterior over $\{d\alpha/dT,\ \alpha_0,\ c,\ d,\ \tau\}$, with temperature scenarios constructed by merging historical Berkeley Earth observations with IPCC SSP projections (shifted to the calibration baseline).

2.3 Section 3: Quadratic Extrapolation of Observations (Cell 8)

As a purely kinematic benchmark, GMSL is extrapolated forward using the rate and acceleration estimated at the end of the Frederikse record via a 30-year bandwidth local polynomial:

$$H(t) = H_{\text{end}} + \dot{H}_{\text{end}}\,(t - t_{\text{end}}) + \tfrac{1}{2}\,\ddot{H}_{\text{end}}\,(t - t_{\text{end}})^2. \tag{3}$$

At $t_{\text{end}} \approx 2003$: rate $= 2.98 \pm 0.10\,\mathrm{mm\,yr^{-1}}$, acceleration $= 0.105 \pm 0.018\,\mathrm{mm\,yr^{-2}}$. A separate extrapolation from mid-record NASA satellite GMSL (anchor ≈ 2009) gives rate $= 3.31\,\mathrm{mm\,yr^{-1}}$, acceleration $= 0.075\,\mathrm{mm\,yr^{-2}}$.

Uncertainty is propagated by bootstrap resampling of rate and acceleration from their Gaussian uncertainties (10^5 draws).

2.4 Section 4: Observed Time Series (Cell 10)

Figure 1 shows Frederikse GMSL (cm, left axis) and 10-year smoothed Berkeley Earth GMST (°C, right axis) on a common timeline, both rebased to year 2000.

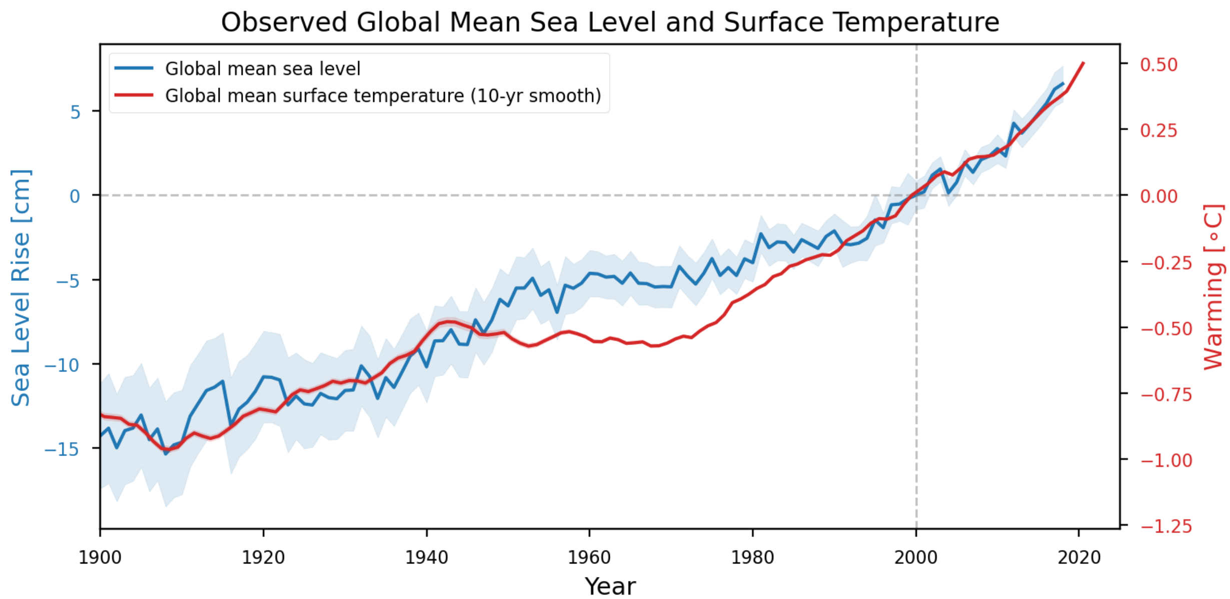

Figure 1: Observed global mean sea level (Frederikse et al., blue, left axis) and global mean surface temperature (Berkeley Earth 10-year Lowess smooth, red, right axis), both rebased to year 2000. The co-evolution of GMSL and GMST motivates the semi-empirical modelling approach.

2.5 Section 5: Two-Panel Projection Plot (Cell 12)

Figure 2 is a two-panel figure:

- **Upper panel:** GMSL projections for SSP2-4.5 from all three sources (observations + kinematic extrapolation, semi-empirical model, IPCC AR6) with 90% confidence intervals and right-axis societal impact scales (people at risk from Kulp & Strauss 2019; annual flood costs from Jevrejeva et al. 2018).
- **Lower panel:** Observed and projected GMST for the same SSP.

2.6 Section 6: Hindcast Cross-Validation (Cells 14–18)

The Bayesian model is re-calibrated on truncated observational records ending at cutoff years 2000, 2010, and 2018. For each cutoff, the model projects forward using observed Berkeley Earth temperatures merged with SSP2-4.5.

Figure 3 shows the hindcast projections with 90% CIs overlaid on the full Frederikse record.

Figure 4 adds the IPCC AR6 workflow ensembles (7 workflows, each with 20,000 MC samples) to the hindcast comparison.

Figure 5 compares NASA satellite GMSL, Frederikse tide-gauge GMSL, a mid-record kinematic extrapolation, and all 7 IPCC AR6 workflows.

Hindcast metrics (vs. Frederikse, 90% CI):

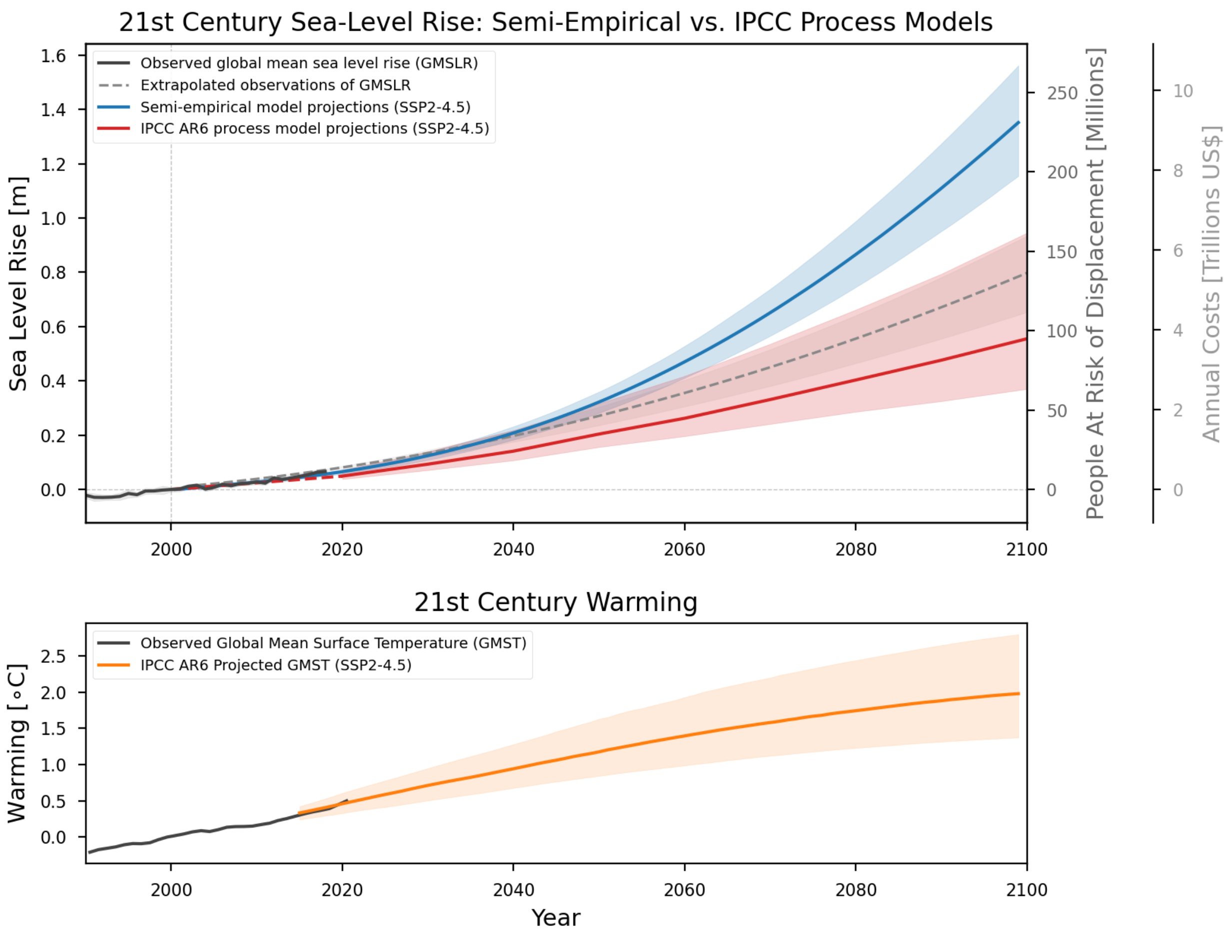

Figure 2: 21st-century projections under SSP2-4.5. **Upper:** GMSL from observations (dark grey), kinematic extrapolation (dashed grey), semi-empirical rate-and-state model (blue), and IPCC AR6 process models (red), with 90% confidence intervals. Right axes show people at risk of displacement (Kulp & Strauss, 2019) and annual flood costs (Jevrejeva et al., 2018). **Lower:** Observed and projected GMST for the same scenario.

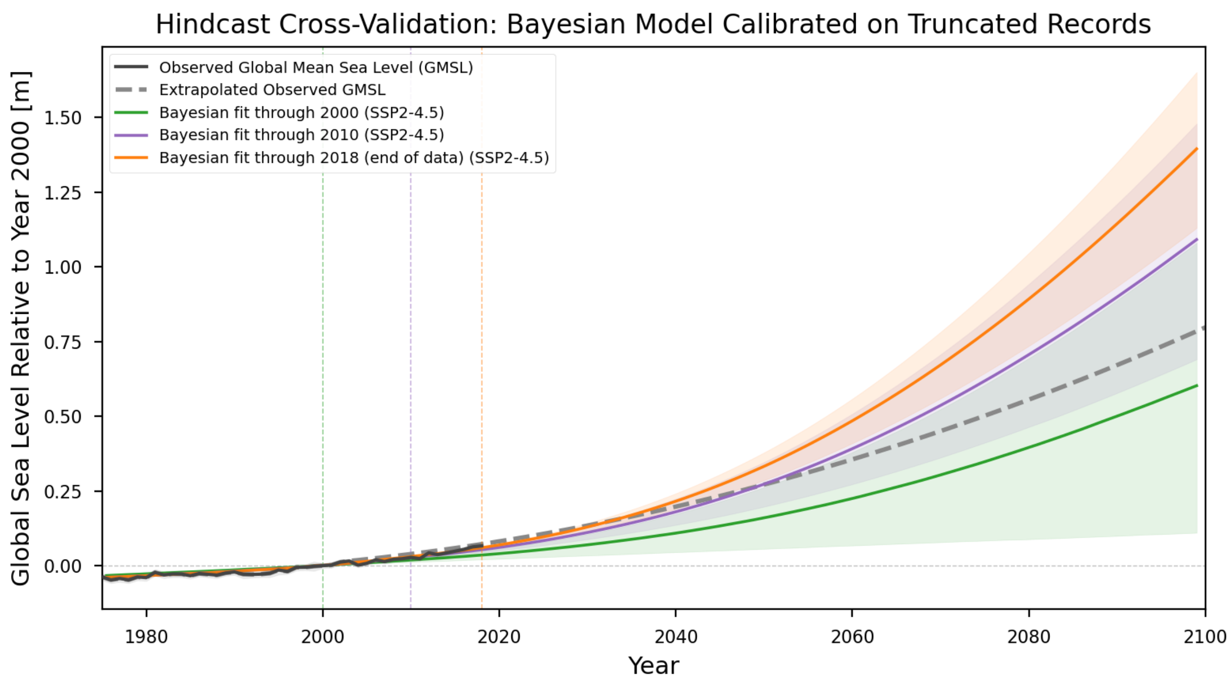

Figure 3: Hindcast cross-validation. The Bayesian rate-and-state model is calibrated on truncated Frederikse GMSL records ending at 2000 (green), 2010 (purple), and 2018 (orange), then projected forward using observed + SSP2-4.5 temperature. Shaded regions show 90% confidence intervals. The kinematic extrapolation (dashed grey) is shown for comparison. Shorter calibration records produce wider uncertainty and larger bias.

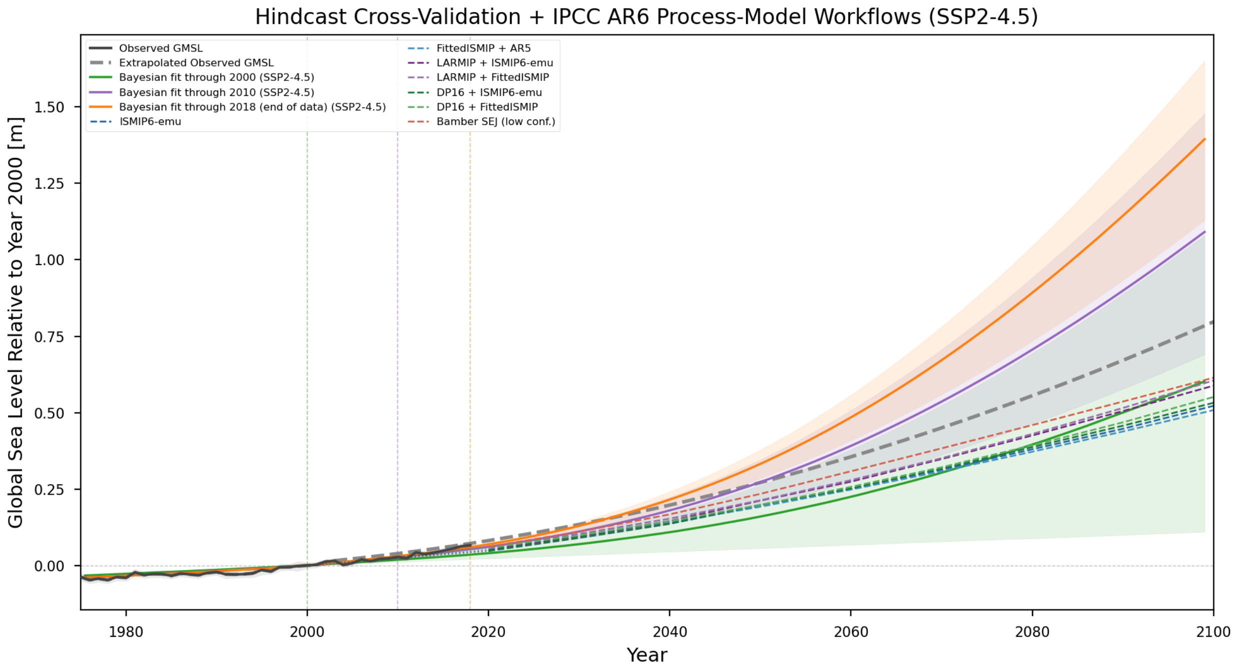

Figure 4: Hindcast cross-validation with IPCC AR6 workflow ensembles overlaid. Seven IPCC workflows (colour-coded) are shown alongside the semi-empirical hindcasts from Figure 3. IPCC workflow medians at 2100 (SSP2-4.5) range from 508 mm (wf_1f) to 614 mm (wf_4, Bamber SEJ, low confidence).

Cutoff	RMSE (mm)	Bias (mm)	90% Coverage
2000	14.8	+10.7	22%
2010	8.6	+5.7	38%
2018	— (end of data)	—	—

Projection comparison at 2100 (SSP2-4.5):

Source	Median (mm)	90% CI (mm)
Rate-and-state	1351	1155–1562
IPCC AR6	555	370–945
Difference	+796	—

2.7 Section 7: Forecasting Skill Analysis (Cells 20–24)

This section performs a rigorous out-of-sample verification of all three model families against NASA satellite altimetry over 2006–2024, using three complementary diagnostics.

2.7.1 Vermeer & Rahmstorf (2009) Implementation (Cell 21)

The VR09 dual model is:

$$\frac{dH}{dt} = a\,(T - T_0) + b\,\frac{dT}{dt}, \tag{4}$$

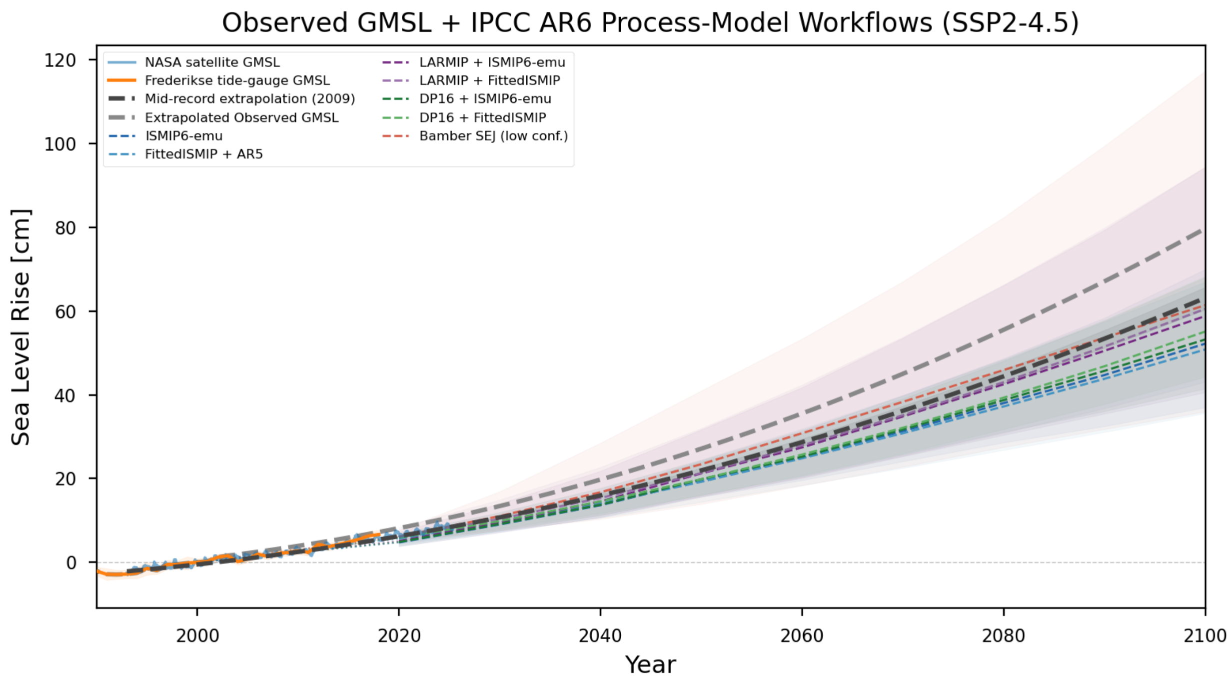

Figure 5: Observed GMSL from NASA satellite altimetry and Frederikse tide-gauge reconstruction, with mid-record kinematic extrapolation from the satellite era and IPCC AR6 workflow ensemble projections. This figure highlights the relationship between observational trends and the spread of process-model projections.

with published parameters $a = 5.6 \pm 0.5\,\mathrm{mm\,yr^{-1}\,K^{-1}}$, $b = -49 \pm 10\,\mathrm{mm\,K^{-1}}$. The equilibrium temperature T_0 is baseline-dependent; two approaches are tested:

- **Approach A** (analytical baseline shift): $T_0 = -0.959\,\mathrm{K}$. RMSE $= 35.2\,\mathrm{mm}$, bias $= -24.6\,\mathrm{mm}$.
- **Approach B** (empirical T_0 calibration with a, b fixed): $T_0 = -0.864\,\mathrm{K}$. RMSE $= 21.2\,\mathrm{mm}$, bias $= -3.1\,\mathrm{mm}$.

Approach B is adopted. An MC ensemble ($N = 2000$) samples jointly over (a, b, T_0) uncertainties.

2.7.2 Direct Out-of-Sample Verification (Cell 22)

Figure 6 compares all models against NASA satellite GMSL (2006–2024).

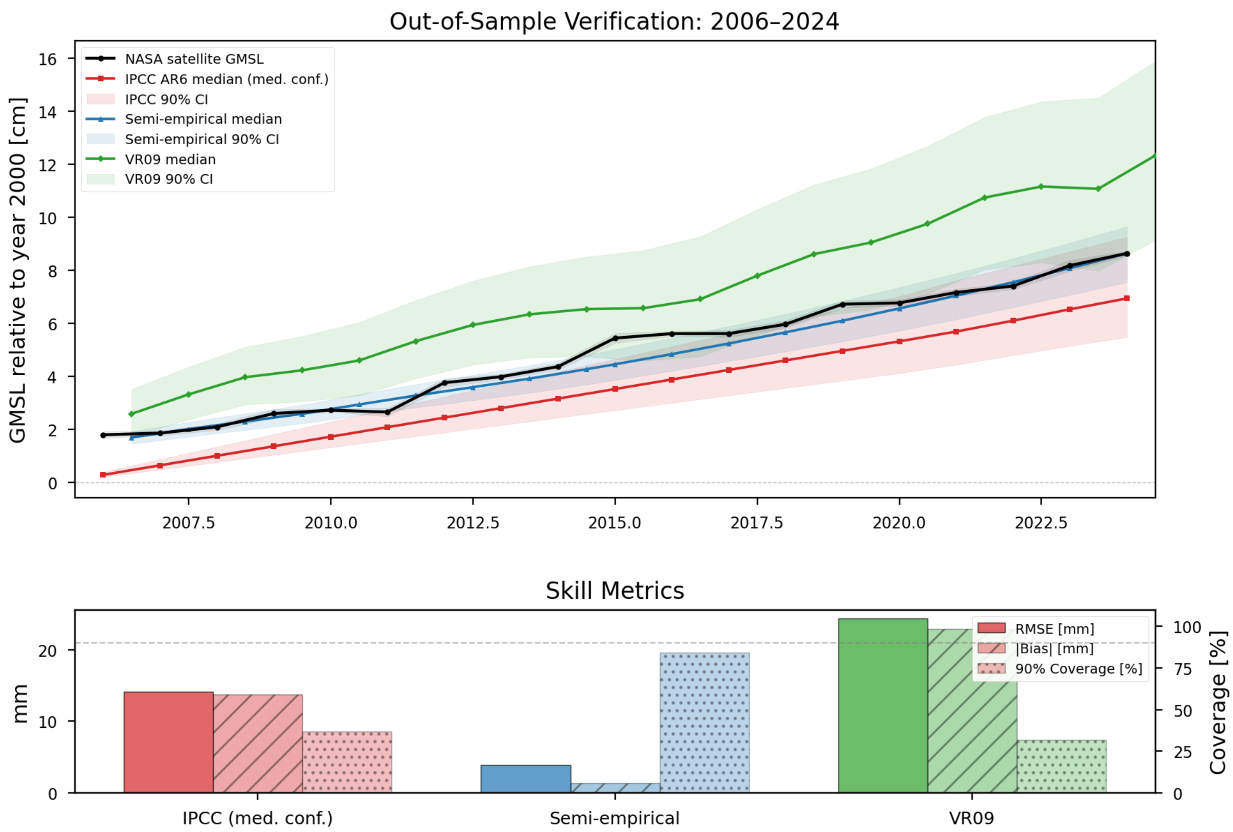

Figure 6: Direct out-of-sample verification against NASA satellite GMSL (2006–2024). **Upper:** Time series comparison of observed GMSL (black) with the semi-empirical model (blue), IPCC AR6 combined medium-confidence ensemble (red), and VR09 (green), each with 90% confidence intervals. **Lower:** Bar chart of RMSE, bias, and 90% CI coverage for all IPCC workflows, the semi-empirical model, and VR09. The semi-empirical model achieves the lowest RMSE (3.9 mm) and highest coverage (84%).

Model	RMSE (mm)	Bias (mm)	90% Coverage
Semi-empirical (DOLS)	3.87	+1.37	84%
IPCC combined (med. conf.)	14.07	+13.73	37%
VR09	24.35	−22.90	32%
Individual IPCC workflows:			
Bamber SEJ (wf_4, low conf.)	8.74	+8.13	63%
LARMIP + FittedISMIP (wf_2f)	10.08	+9.69	74%
ISMIP6-emu (wf_1e)	17.82	+17.32	0%

The DOLS model has the lowest RMSE, smallest bias, and highest CI coverage among all models tested. IPCC workflows systematically overpredict early-period GMSL (positive bias), while VR09 strongly underpredicts (negative bias).

2.7.3 Reliability Analysis (Cell 23)

Figure 7 presents probability integral transform (PIT) histograms and reliability (coverage) diagrams for the three model families.

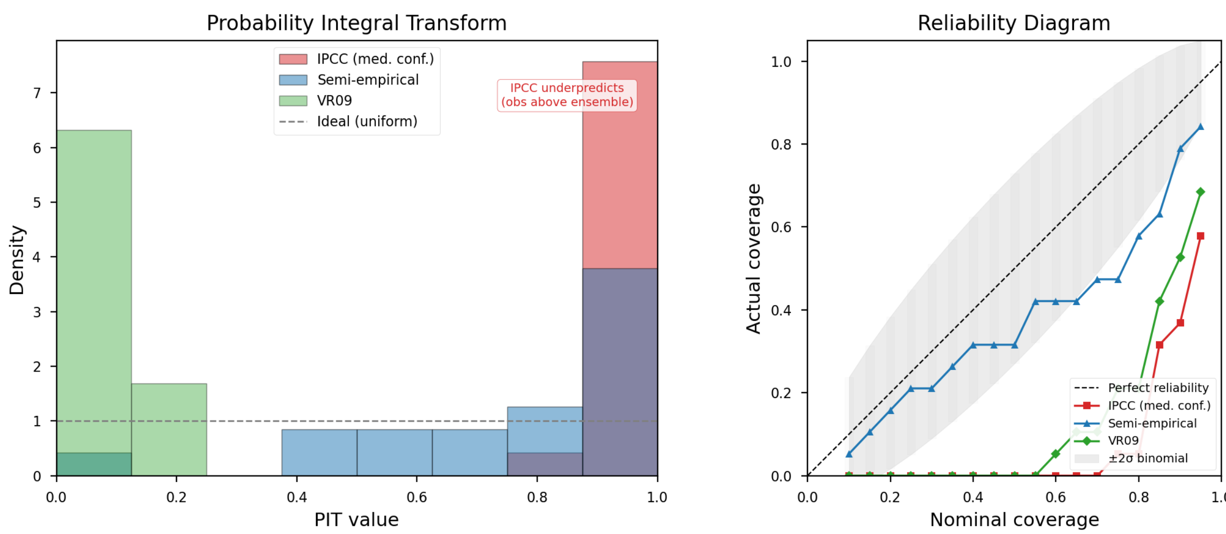

Figure 7: Reliability analysis. **Left:** PIT histograms for the IPCC combined ensemble, the semi-empirical model (DOLS), and VR09. A well-calibrated model produces a uniform histogram. The IPCC histogram is concentrated near 1.0 (systematic underprediction of observed GMSL); VR09 is concentrated near 0.0 (systematic overprediction); DOLS is closest to uniform but skewed toward high values. **Right:** Reliability (coverage) diagrams showing actual vs. nominal coverage at each confidence level. The diagonal represents a perfectly calibrated model.

The PIT is the CDF value at which the observation falls within each model's ensemble distribution at each verification year. A well-calibrated model produces PIT values uniformly distributed on $[0, 1]$.

Model	PIT Mean	KS Statistic	p-value	90% Coverage
IPCC	0.956	0.875	< 0.001	37%
DOLS	0.762	0.412	0.002	79%
VR09	0.072	0.778	< 0.001	53%

All models fail the KS uniformity test, but the DOLS model is closest to calibrated (PIT mean = 0.762 vs. the ideal 0.50). The IPCC PIT mean near 1.0 confirms systematic underprediction; VR09's near-zero PIT mean confirms systematic overprediction.

2.7.4 Conditional Skill Analysis (Cell 24)

Figure 8 investigates whether skill differences arise from temperature forcing uncertainty or from the GMSL model itself. The DOLS model is re-evaluated using observed Berkeley Earth GMST only (no SSP switch at end of record), isolating the sea-level model's skill from temperature scenario uncertainty.

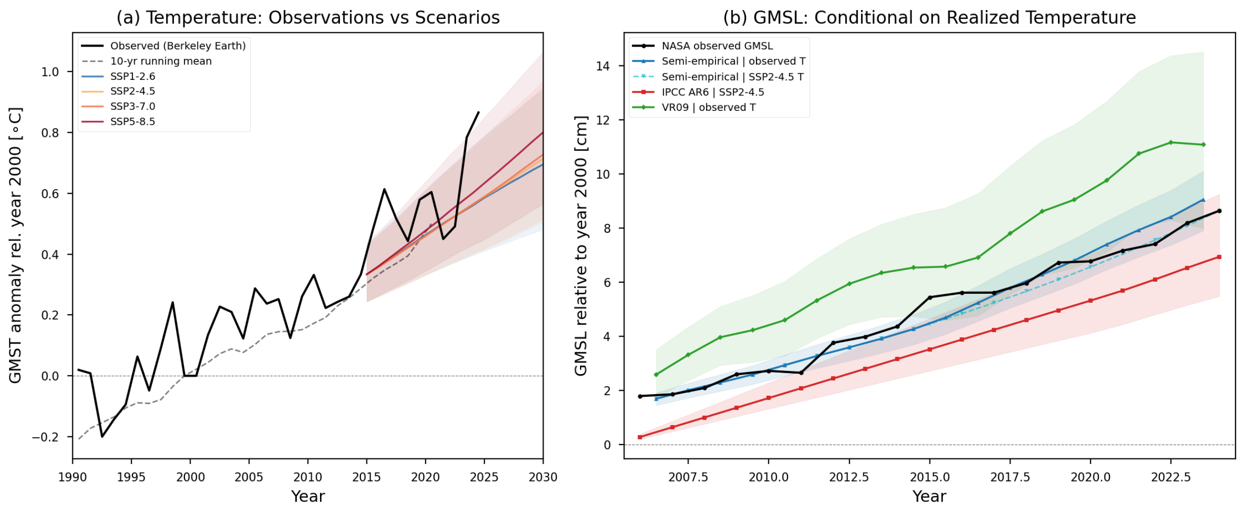

Figure 8: Conditional skill analysis. **Left:** Observed GMST (Berkeley Earth) and SSP2-4.5 projected temperature over the verification period, showing that realized warming closely tracks the SSP2-4.5 scenario through 2024. **Right:** GMSL projections conditioned on observed temperature only (no SSP switch), with 90% CIs. The DOLS model conditioned on observed temperature achieves RMSE = 5.5 mm and 89% coverage, confirming that its skill derives from the sea-level model rather than temperature scenario accuracy.

Configuration	RMSE (mm)	Bias (mm)	90% Coverage
DOLS | observed T	5.46	−2.47	89%
DOLS | SSP2-4.5 T	3.87	+1.37	—
IPCC | SSP2-4.5	14.07	+13.73	37%
VR09 | observed T	23.62	−22.35	37%

When conditioned on observed temperature, the DOLS model achieves RMSE = 5.5 mm with 89% coverage at the 90% nominal level. The IPCC bias is not attributable to temperature forcing because SSP2-4.5 closely tracks observed warming through 2024.

2.8 Section 8: Forecast Horizon Analysis (Cells 26–28)

Three diagnostics assess how model skill degrades with lead time and extrapolation beyond the calibration domain.

2.8.1 Hindcast Degradation Curves (Cell 26)

Figure 9 re-calibrates both DOLS and VR09 for cutoff years from 1960 to 2010 (in 10-year steps), projects forward with observed temperature only, and bins RMSE and 90% CI coverage by 5-year lead time.

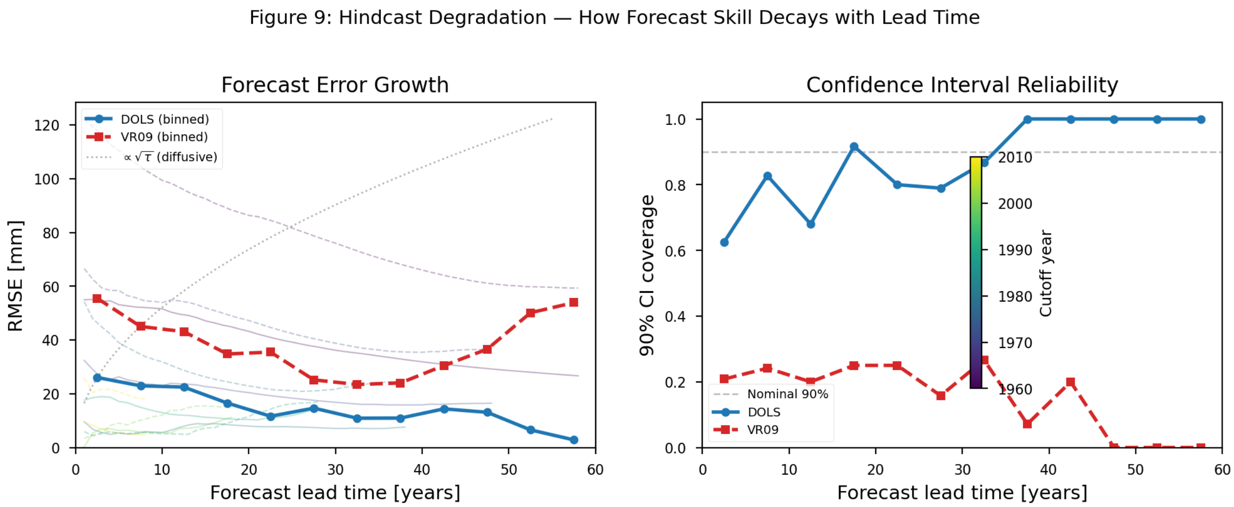

Figure 9: Hindcast degradation curves. RMSE (left) and 90% CI coverage (right) as a function of forecast lead time for the semi-empirical model (DOLS, blue) and VR09 (green), computed by re-calibrating on records truncated at 1960–2010 and verifying against subsequent observations. DOLS error remains below ∼27 mm at all leads and coverage stays at or above 62%. VR09 error grows steadily and coverage falls to 0% at leads exceeding 35 years.

Key findings:

- DOLS RMSE remains below ∼27 mm at all lead times up to 58 years and *decreases* at longer leads (most recent data has highest information content). Coverage stays at or above 62%, reaching 100% at leads > 35 years.
- VR09 RMSE grows to ∼54 mm at 58-year lead and coverage falls to 0%.

2.8.2 Temperature Exceedance Analysis (Cell 27)

Figure 10 quantifies how far projected temperatures exceed the maximum observed during calibration.

The full calibration record (1920–2018) has $T_{\max} = 0.492\,^{\circ}\mathrm{C}$ (Berkeley Earth 1995–2005 baseline). All SSPs enter the temperature extrapolation regime around 2025–2027:

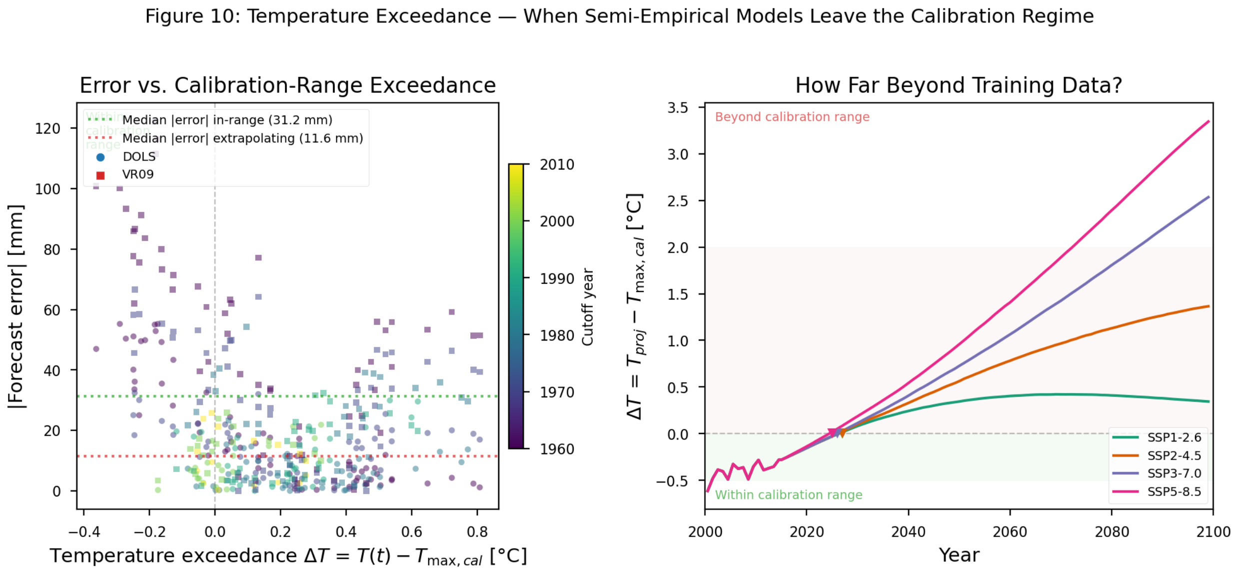

Figure 10: Temperature exceedance analysis. Projected GMST under each SSP relative to the maximum temperature in the calibration record ($T_{\max} = 0.49\,°C$, Berkeley Earth 1995–2005 baseline). All SSPs enter the extrapolation regime ($T > T_{\max}$) around 2025–2027. The semi-empirical model's absolute error is plotted against temperature exceedance, showing that errors in the extrapolation regime are not systematically larger than errors within the calibration range.

SSP	Exceedance Year	ΔT at 2100 (°C)
SSP1-2.6	2027	+0.34
SSP2-4.5	2027	+1.36
SSP3-7.0	2026	+2.53
SSP5-8.5	2025	+3.34

Counterintuitively, the median absolute error is *smaller* in the extrapolation regime (11.6 mm, $n = 298$) than within the calibration range (31.2 mm, $n = 98$), likely because recent (warmer) data is most informative for near-future projections.

2.8.3 Cross-Model Divergence (Cell 28)

Figure 11 compares DOLS, VR09, kinematic extrapolation, and IPCC AR6 on a common grid (2000–2100, SSP2-4.5). The inter-model standard deviation (σ_{inter}) is compared to the mean intra-model uncertainty (σ_{intra}).

Year	DOLS (mm)	VR09 (mm)	Kinematic (mm)	IPCC (mm)	σ_{inter} (mm)	$\sigma_{\text{inter}}/\sigma_{\text{intra}}$
2030	125	178	134	95	30	1.28
2050	322	344	271	200	55	1.08
2100	1351	954	797	546	291	1.62

The inter-model spread exceeds the intra-model uncertainty at all times ($\sigma_{\text{inter}}/\sigma_{\text{intra}} \geq 1$), indicating that structural model differences dominate over parametric uncertainty. At 2100, the divergence ratio reaches 1.62.

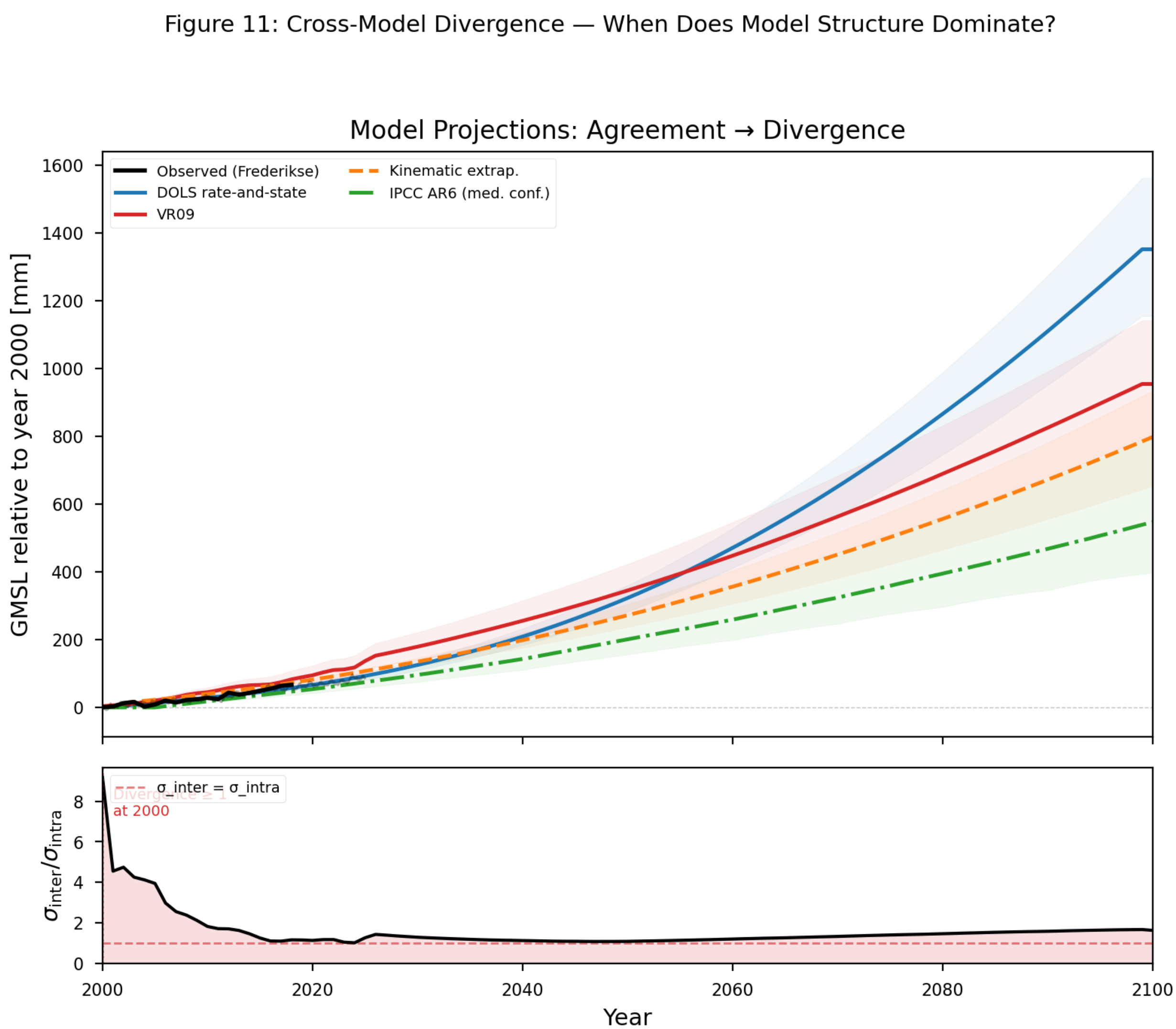

Figure 11: Cross-model divergence under SSP2-4.5. **Upper:** GMSL projections from the semi-empirical model (DOLS), VR09, kinematic extrapolation, and IPCC AR6, each with 90% confidence intervals. **Lower:** Ratio of inter-model standard deviation (σ_{inter}) to mean intra-model uncertainty (σ_{intra}). A ratio exceeding 1.0 indicates that structural model differences dominate over parametric uncertainty within any single model. The ratio is ≥ 1 at all projection times and reaches 1.62 at 2100.

3 Key Figures

1. **Figure 1:** Observed GMSL + GMST twin-axis time series (1900–2024).
2. **Figure 2:** Two-panel GMSL/GMST projection with societal impact axes.
3. **Figure 3:** Hindcast cross-validation (DOLS at cutoffs 2000, 2010, 2018).
4. **Figure 4:** Hindcast cross-validation with IPCC AR6 workflow ensembles overlaid.
5. **Figure 5:** NASA satellite GMSL + Frederikse + kinematic extrapolation + IPCC AR6 workflows.
6. **Figure 6:** Direct out-of-sample verification (2006–2024) with skill metrics.
7. **Figure 7:** PIT histograms and reliability diagrams.
8. **Figure 8:** Conditional skill analysis (observed vs. SSP temperature).
9. **Figure 9:** Hindcast degradation curves (RMSE and coverage vs. lead time).
10. **Figure 10:** Temperature exceedance analysis.
11. **Figure 11:** Cross-model divergence (σ_{inter} vs. σ_{intra}).

4 Dependencies

The notebook imports from the following project modules:

- `slr_analysis.py` — DOLS calibration (`calibrate_dols`), kinematics (`compute_kinematics`), resampling
- `slr_data_readers.py` — societal impact functions (`people_displaced_kulpstrauss2019`, `slr_cost_jevrejeva2018`)
- `slr_projections.py` — MC ensemble projections (`project_gmsl_state_ensemble`)
- `bayesian_dols.py` — ODE solver and Bayesian fitting (`solve_state_ode`, `fit_bayesian_level`)
- `slr_ar6_member_readers.py` — IPCC AR6 full-sample component readers (224 NetCDF files, 7 workflows)

Data are stored in `data/processed/slr_processed_data.h5` (53 keys) and `data/raw/ipcc_ar6/` (FACTS NetCDF files).

5 Reproducibility

- All random number generators use explicit seeds (42, 99, etc.) for reproducibility.
- Monte Carlo ensemble size: $N = 2{,}000$ (DOLS, VR09); $N = 20{,}000$ per workflow (IPCC); $N = 120{,}000$ combined medium-confidence.
- Verification window: NASA satellite GMSL, 2006–2024 (19 annual values).
- IPCC workflows interpolated from decadal to annual resolution for year-by-year comparison.